

# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1     | 2      | 1      |
| -1    | $x_2$ | 0     | 4      | 1      |
| 1     | $x_3$ | 0     | 3      | 0      |
| -1    | $x_4$ | 0     | 3      | 1      |
| 1     | $x_5$ | 5     | 0      | 3      |
| -1    | $x_6$ | 1     | 4      | 3      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|  |  |  |  |  |  |   |
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Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

| $x_i$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   x - x'  ^2}$ | $e^{-\gamma   x - x'  ^2}$ |   |
|-------|--------------|------------------------------------|----------------------------|---|
| 1     |              |                                    |                            | 2 |
| 0     |              |                                    |                            | 4 |
| 0     |              |                                    |                            | 3 |
| 0     |              |                                    |                            |   |
| 5     |              |                                    |                            |   |
| 1     |              |                                    |                            |   |

| $-  \gamma  ^2$ | $e^{\square}$ |
|-----------------|---------------|
| -0.5            | 0.6           |
| -0.4            | 0.67          |
| -0.3            | 0.7           |
| -0.2            | 0.8           |
| -0.1            | 0.9           |

|   |
|---|
| 1 |
| 1 |
| 1 |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

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|  |  |  |  |  |  | -1 |
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# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |

Kernel Matrix

$K(x_i, x'_j)$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|  |  |  |  |  |  |   |
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sign 

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Decision Boundary

$b + \sum a_i y_i K$

## RBF

| $x_i$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   (-0.1)  ^2} e^{-\gamma   x - x'  ^2}$ | $x'_j$ | $-  \gamma  ^2$ | $e^{-\gamma   x - x'  ^2}$ |   |
|-------|--------------|-------------------------------------------------------------|--------|-----------------|----------------------------|---|
| 1 2 1 |              |                                                             | 2      | -0.5            | 0.6                        | 1 |
| 0 4 3 |              |                                                             | 4      | -0.4            | 0.67                       | 1 |
| 0 3 0 |              |                                                             | 3      | -0.3            | 0.7                        | 1 |
| 0 3 1 |              |                                                             |        | -0.2            | 0.8                        |   |
| 5 0 3 |              |                                                             |        | -0.1            | 0.9                        |   |
| 1 4 3 |              |                                                             |        |                 |                            |   |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |  |    |
|--|--|--|--|--|--|----|
|  |  |  |  |  |  | -1 |
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sign 

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# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |

Kernel Matrix

$K(x_i, x'_j)$

$a_i$

$b$

$a_i y_i$

sign

Decision Boundary

$b + \sum a_i y_i K$

## RBF

| $x_i$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   (-0.1)  ^2} e^{\square}$ |  |
|-------|--------------|------------------------------------------------|--|
| 1 2 1 |              |                                                |  |
| 0 4 3 |              |                                                |  |
| 0 3 0 |              |                                                |  |
| 0 3 1 |              |                                                |  |
| 5 0 3 |              |                                                |  |
| 1 4 3 |              |                                                |  |

$a_i$

$b$

$a_i y_i$

sign

# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 |        |        |
| -1    | $x_4$ | 0     | 3 |        |        |
| 1     | $x_5$ | 5     | 0 |        |        |
| -1    | $x_6$ | 1     | 4 |        |        |

Kernel Matrix

$K(x_i, x'_j)$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| 2 | 0 | 1 | -1 | -1 | -1 |
|---|---|---|----|----|----|

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sign 

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Decision Boundary

$b + \sum a_i y_i K$

## RBF

| $x_i$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   (-0.1)  ^2} e^{-\gamma   x - x'  ^2}$ | $x'_j$ | $-  \gamma  ^2$ | $e^{-\gamma   x - x'  ^2}$ |   |
|-------|--------------|-------------------------------------------------------------|--------|-----------------|----------------------------|---|
| 1     |              |                                                             | 2      | -0.5            | 0.6                        | 1 |
| 0     |              |                                                             | 4      | -0.4            | 0.67                       | 1 |
| 0     |              |                                                             | 3      | -0.3            | 0.7                        | 1 |
| 0     |              |                                                             |        | -0.2            | 0.8                        |   |
| 5     |              |                                                             |        | -0.1            | 0.9                        |   |
| 1     |              |                                                             |        |                 |                            |   |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

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# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 | 0      | 1      |
| -1    | $x_4$ | 0     | 3 | 1      | 1      |
| 1     | $x_5$ | 5     | 0 | 3      | 1      |
| -1    | $x_6$ | 1     | 4 | 3      | 1      |

Kernel Matrix

$K(x_i, x'_j)$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
|---|---|---|----|---|----|

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Decision Boundary

$b + \sum a_i y_i K$

## RBF

| $x_i$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   (-0.1)  ^2} e^{\square}$ | $x'_j$ | $-  \gamma  ^2$ | $e^{\square}$ |   |
|-------|--------------|------------------------------------------------|--------|-----------------|---------------|---|
| 1     |              |                                                | 2      | -0.5            | 0.6           | 1 |
| 0     |              |                                                | 4      | -0.4            | 0.67          | 1 |
| 0     |              |                                                | 3      | -0.3            | 0.7           | 1 |
| 0     |              |                                                |        | -0.2            | 0.8           |   |
| 5     |              |                                                |        | -0.1            | 0.9           |   |
| 1     |              |                                                |        |                 |               |   |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
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# SVM

Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |

Kernel Matrix

$K(x_i, x'_j)$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
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Decision Boundary

$b + \sum a_i y_i K$

RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $(-0.1) e^\square$

|   |    |   |
|---|----|---|
| 1 | 4  | 4 |
| 4 | 0  | 0 |
| 0 | 1  | 0 |
| 4 | 1  | 4 |
| 9 | 16 | 0 |
| 1 | 0  | 0 |

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$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
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sign 

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# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 |        |        |
| -1    | $x_4$ | 0     | 3 |        |        |
| 1     | $x_5$ | 5     | 0 |        |        |
| -1    | $x_6$ | 1     | 4 |        |        |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
|---|---|---|----|---|----|

 $\rightarrow$ 

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|----|---|
| 17 | 5 |
|----|---|

sign 

|   |   |
|---|---|
| + | + |
|---|---|

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $e^{-0.1}$

|   |    |   |    |  |  |
|---|----|---|----|--|--|
| 1 | 4  | 4 | 9  |  |  |
| 4 | 0  | 0 | 4  |  |  |
| 0 | 1  | 0 | 1  |  |  |
| 4 | 1  | 4 | 9  |  |  |
| 9 | 16 | 0 | 25 |  |  |
| 1 | 0  | 0 | 1  |  |  |

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$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
|--|--|--|--|--|----|

sign 

|  |
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|  |
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# Linear

|        | $x'_1$ | $x'_2$ |
|--------|--------|--------|
| $x'_j$ | 2      | 1      |
|        | 4      | 1      |
|        | 3      | 1      |

$$K(\mathbf{x}_i, \mathbf{x}'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$ 

|   |   |   |    |    |    |   |
|---|---|---|----|----|----|---|
| 2 | 0 | 1 | -1 | -1 | -1 | 2 |
|---|---|---|----|----|----|---|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

  
 $a_i y_i$ 

|   |   |   |    |    |    |   |
|---|---|---|----|----|----|---|
| 2 | 0 | 1 | -1 | -1 | -1 | 2 |
|---|---|---|----|----|----|---|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

  
 $sign$ 

|   |   |
|---|---|
| + | + |
|---|---|

$$\mathbf{b} + \sum \mathbf{a}_i y_i \mathbf{K}$$

**RBF**

Diagram illustrating the RBF (Radial Basis Function) kernel calculation. The input matrix  $X_i$  is a 6x3 matrix. The target vector  $x'_j$  is a 3x1 vector. The L2 distance is calculated as  $\|x - x'\|^2$ . The kernel function is  $\exp(-\gamma \|x - x'\|^2)$ , where  $\gamma = 0.1$ . The output matrix is a 6x3 matrix.

Input Matrix  $X_i$ :

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

Target Vector  $x'_j$ :

|   |
|---|
| 2 |
| 4 |
| 3 |

Output Matrix:

|   |    |   |
|---|----|---|
| 1 | 4  | 4 |
| 4 | 0  | 0 |
| 0 | 1  | 0 |
| 4 | 1  | 4 |
| 9 | 16 | 0 |
| 1 | 0  | 0 |

Kernel Function Results (Intermediate Steps):

|    |   |  |
|----|---|--|
| 9  | 3 |  |
| 4  | 2 |  |
| 1  | 1 |  |
| 9  | 3 |  |
| 25 | 5 |  |
| 1  | 1 |  |

Final Output Matrix (Green):

|  |  |  |
|--|--|--|
|  |  |  |
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|  |  |  |
|  |  |  |
|  |  |  |

|   |
|---|
| 1 |
| 1 |
| 1 |

|           |    |    |    |    |   |    |      |
|-----------|----|----|----|----|---|----|------|
| $a_i$     | 20 | 10 | 10 | 10 | 0 | 10 | $b$  |
| $a_i y_i$ |    |    |    |    |   |    | -1   |
|           |    |    |    |    |   |    | sign |

# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
|---|---|---|----|---|----|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
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sign 

|   |   |
|---|---|
| + | + |
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Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $(-0.1) e^\square$

|   |    |   |    |   |      |
|---|----|---|----|---|------|
| 1 | 4  | 4 | 9  | 3 | -0.3 |
| 4 | 0  | 0 | 4  | 2 | -0.2 |
| 0 | 1  | 0 | 1  | 1 | -0.1 |
| 4 | 1  | 4 | 9  | 3 | -0.3 |
| 9 | 16 | 0 | 25 | 5 | -0.5 |
| 1 | 0  | 0 | 1  | 1 | -0.1 |

| $-\gamma   ^2$ | $e^\square$ |   |
|----------------|-------------|---|
| -0.5           | 0.6         | 1 |
| -0.4           | 0.67        | 1 |
| -0.3           | 0.7         | 1 |
| -0.2           | 0.8         |   |
| -0.1           | 0.9         |   |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
|--|--|--|--|--|----|

sign 

|  |
|--|
|  |
|--|

# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
|---|---|---|----|---|----|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

sign 

|   |   |
|---|---|
| + | + |
|---|---|

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $(-0.1) e^\square$

|   |    |   |    |   |      |     |
|---|----|---|----|---|------|-----|
| 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

| $-\gamma   ^2$ | $e^\square$ |
|----------------|-------------|
| -0.5           | 0.6         |
| -0.4           | 0.67        |
| -0.3           | 0.7         |
| -0.2           | 0.8         |
| -0.1           | 0.9         |

|   |
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| 1 |
| 1 |
| 1 |

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|  |  |  |  |  |  |  |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
|--|--|--|--|--|----|

sign 

|  |
|--|
|  |
|--|

# SVM

Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |
|       |       |       | 13     | 2      |
|       |       |       | 25     | 7      |
|       |       |       | 12     | 3      |
|       |       |       | 15     | 4      |
|       |       |       | 19     | 8      |
|       |       |       | 27     | 8      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$  2 0 1 1 1 1  $b$   
 $a_i y_i$  2 0 1 -1 1 -1 2  $\rightarrow$  17 5  
 sign + +

Decision Boundary

$$b + \sum a_i y_i K$$

RBF

| $x_i$ | $x'_j$ |                             |
|-------|--------|-----------------------------|
| 1 2 1 | 2      |                             |
| 0 4 3 | 4      |                             |
| 0 3 0 | 3      |                             |
| 0 3 1 |        |                             |
| 5 0 3 |        |                             |
| 1 4 3 |        |                             |
|       |        | $L2 \text{ distance }   ^2$ |
|       |        | $(x - x')^2$                |
|       |        | $\sum \sqrt{-\gamma   ^2}$  |
|       |        | $(-0.1) e^{\square}$        |
|       |        | 0.7                         |
|       |        | 0.8                         |
|       |        | 0.9                         |
|       |        | 0.7                         |
|       |        | 0.6                         |
|       |        | 0.9                         |

| $-\gamma   ^2$ | $e^{\square}$ |
|----------------|---------------|
| -0.5           | 0.6           |
| -0.4           | 0.67          |
| -0.3           | 0.7           |
| -0.2           | 0.8           |
| -0.1           | 0.9           |

|   |
|---|
| 1 |
| 1 |
| 1 |

|    |   |   |  |  |  |  |
|----|---|---|--|--|--|--|
| 0  | 1 | 0 |  |  |  |  |
| 1  | 9 | 4 |  |  |  |  |
| 1  | 4 | 1 |  |  |  |  |
| 1  | 4 | 0 |  |  |  |  |
| 16 | 1 | 4 |  |  |  |  |
| 0  | 9 | 4 |  |  |  |  |

$a_i$  20 10 10 10 0 10  $b$   
 $a_i y_i$  -1 -1  
 sign

# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 | 0      | 1      |
| -1    | $x_4$ | 0     | 3 | 1      | 1      |
| 1     | $x_5$ | 5     | 0 | 3      | 1      |
| -1    | $x_6$ | 1     | 4 | 3      | 1      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| 2 | 0 | 1 | -1 | -1 | -1 |
|---|---|---|----|----|----|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

sign 

|   |   |
|---|---|
| + | + |
|---|---|

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $(-0.1) e^\square$

|   |    |   |    |   |      |     |
|---|----|---|----|---|------|-----|
| 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

| $-\gamma   ^2$ | $e^\square$ |   |
|----------------|-------------|---|
| -0.5           | 0.6         | 1 |
| -0.4           | 0.67        | 1 |
| -0.3           | 0.7         | 1 |
| -0.2           | 0.8         |   |
| -0.1           | 0.9         |   |

|    |   |   |    |  |  |  |
|----|---|---|----|--|--|--|
| 0  | 1 | 0 | 1  |  |  |  |
| 1  | 9 | 4 | 14 |  |  |  |
| 1  | 4 | 1 | 6  |  |  |  |
| 1  | 4 | 0 | 5  |  |  |  |
| 16 | 1 | 4 | 21 |  |  |  |
| 0  | 9 | 4 | 13 |  |  |  |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
|--|--|--|--|--|----|

sign 

|  |
|--|
|  |
|--|



# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 | 0      | 1      |
| -1    | $x_4$ | 0     | 3 | 1      | 1      |
| 1     | $x_5$ | 5     | 0 | 3      | 1      |
| -1    | $x_6$ | 1     | 4 | 3      | 1      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$  2 0 1 1 1 1  $b$   
 $a_i y_i$  2 0 1 -1 -1 -1 2  $\rightarrow$  17 5  
 sign + +

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$   $x'_j$   
 L2 distance  $\|x - x'\|^2$   
 $(x - x')^2 \sum \sqrt{-\gamma \|x - x'\|^2} e^{-\gamma \|x - x'\|^2}$   
 $(-0.1) e^{-0.1}$

|   |   |   |   |    |   |    |   |      |     |
|---|---|---|---|----|---|----|---|------|-----|
| 1 | 2 | 1 | 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 0 | 4 | 3 | 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 3 | 0 | 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 0 | 3 | 1 | 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 5 | 0 | 3 | 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 4 | 3 | 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

| $-\gamma \ x - x'\ ^2$ | $e^{-\gamma \ x - x'\ ^2}$ |
|------------------------|----------------------------|
| -0.5                   | 0.6                        |
| -0.4                   | 0.67                       |
| -0.3                   | 0.7                        |
| -0.2                   | 0.8                        |
| -0.1                   | 0.9                        |

|   |
|---|
| 1 |
| 1 |
| 1 |

|    |   |   |    |   |  |  |
|----|---|---|----|---|--|--|
| 0  | 1 | 0 | 1  | 1 |  |  |
| 1  | 9 | 4 | 14 | 3 |  |  |
| 1  | 4 | 1 | 6  | 2 |  |  |
| 1  | 4 | 0 | 5  | 2 |  |  |
| 16 | 1 | 4 | 21 | 4 |  |  |
| 0  | 9 | 4 | 13 | 3 |  |  |

$a_i$  20 10 10 10 0 10  $b$   
 $a_i y_i$  -1 -1  
 sign + +

# SVM

Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 2      | 1      |
| -1    | $x_2$ | 0 4 3 | 4      | 1      |
| 1     | $x_3$ | 0 3 0 | 3      | 1      |
| -1    | $x_4$ | 0 3 1 |        |        |
| 1     | $x_5$ | 5 0 3 |        |        |
| -1    | $x_6$ | 1 4 3 |        |        |
|       |       |       | 13     | 2      |
|       |       |       | 25     | 7      |
|       |       |       | 12     | 3      |
|       |       |       | 15     | 4      |
|       |       |       | 19     | 8      |
|       |       |       | 27     | 8      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$  2 0 1 1 1 1  $b$   
 $a_i y_i$  2 0 1 -1 -1 -1 2  $\rightarrow$  17 5  
 sign + +

Decision Boundary

$$b + \sum a_i y_i K$$

RBF

| $x_i$ | $x'_j$ |                             |
|-------|--------|-----------------------------|
| 1 2 1 | 2      |                             |
| 0 4 3 | 4      |                             |
| 0 3 0 | 3      |                             |
| 0 3 1 |        |                             |
| 5 0 3 |        |                             |
| 1 4 3 |        |                             |
|       |        | $L2 \text{ distance }   ^2$ |
|       |        | $(x - x')^2$                |
|       |        | $\sum \sqrt{-\gamma   ^2}$  |
|       |        | $e^{-0.1}$                  |
|       |        | 0.7                         |
|       |        | 0.8                         |
|       |        | 0.9                         |
|       |        | 0.7                         |
|       |        | 0.6                         |
|       |        | 0.9                         |

| $-\gamma   ^2$ | $e^{-\gamma   ^2}$ |
|----------------|--------------------|
| -0.5           | 0.6                |
| -0.4           | 0.67               |
| -0.3           | 0.7                |
| -0.2           | 0.8                |
| -0.1           | 0.9                |

|   |
|---|
| 1 |
| 1 |
| 1 |

|        |    |      |  |
|--------|----|------|--|
| 0 1 0  | 1  | -0.1 |  |
| 1 9 4  | 14 | -0.3 |  |
| 1 4 1  | 6  | -0.2 |  |
| 1 4 0  | 5  | -0.2 |  |
| 16 1 4 | 21 | -0.4 |  |
| 0 9 4  | 13 | -0.3 |  |

$a_i$  20 10 10 10 0 10  $b$   
 $a_i y_i$  -1   
 sign

# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 |        |        |
| -1    | $x_4$ | 0     | 3 |        |        |
| 1     | $x_5$ | 5     | 0 |        |        |
| -1    | $x_6$ | 1     | 4 |        |        |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| 2 | 0 | 1 | -1 | -1 | -1 |
|---|---|---|----|----|----|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

sign 

|   |   |
|---|---|
| + | + |
|---|---|

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-y||^2}$   
 $e^{-0.1}$

|   |    |   |    |   |      |     |
|---|----|---|----|---|------|-----|
| 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

| $-y  ^2$ | $e^{-}$ |   |
|----------|---------|---|
| -0.5     | 0.6     | 1 |
| -0.4     | 0.67    | 1 |
| -0.3     | 0.7     | 1 |
| -0.2     | 0.8     |   |
| -0.1     | 0.9     |   |

|    |   |   |    |   |      |      |
|----|---|---|----|---|------|------|
| 0  | 1 | 0 | 1  | 1 | -0.1 | 0.9  |
| 1  | 9 | 4 | 14 | 3 | -0.3 | 0.7  |
| 1  | 4 | 1 | 6  | 2 | -0.2 | 0.8  |
| 1  | 4 | 0 | 5  | 2 | -0.2 | 0.8  |
| 16 | 1 | 4 | 21 | 4 | -0.4 | 0.67 |
| 0  | 9 | 4 | 13 | 3 | -0.3 | 0.9  |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|  |  |  |  |  |    |
|--|--|--|--|--|----|
|  |  |  |  |  | -1 |
|--|--|--|--|--|----|

sign 

|  |
|--|
|  |
|--|

# SVM

## Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 |        |        |
| -1    | $x_4$ | 0     | 3 |        |        |
| 1     | $x_5$ | 5     | 0 |        |        |
| -1    | $x_6$ | 1     | 4 |        |        |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|

 $b$

$a_i y_i$ 

|   |   |   |    |   |    |
|---|---|---|----|---|----|
| 2 | 0 | 1 | -1 | 1 | -1 |
|---|---|---|----|---|----|

 $\rightarrow$ 

|    |   |
|----|---|
| 17 | 5 |
|----|---|

sign 

|   |   |
|---|---|
| + | + |
|---|---|

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

$x_i$ 

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 4 | 3 |
| 0 | 3 | 0 |
| 0 | 3 | 1 |
| 5 | 0 | 3 |
| 1 | 4 | 3 |

$x'_j$ 

|   |
|---|
| 2 |
| 4 |
| 3 |

$L2 \text{ distance } ||^2$   
 $(x - x')^2$   
 $\sum \sqrt{-\gamma ||^2}$   
 $e^{-0.1}$

|   |    |   |    |   |      |     |
|---|----|---|----|---|------|-----|
| 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

| $-\gamma   ^2$ | $e^{-\gamma   ^2}$ |   |
|----------------|--------------------|---|
| -0.5           | 0.6                | 1 |
| -0.4           | 0.67               | 1 |
| -0.3           | 0.7                | 1 |
| -0.2           | 0.8                |   |
| -0.1           | 0.9                |   |

|    |   |   |    |   |      |      |
|----|---|---|----|---|------|------|
| 0  | 1 | 0 | 1  | 1 | -0.1 | 0.9  |
| 1  | 9 | 4 | 14 | 3 | -0.3 | 0.7  |
| 1  | 4 | 1 | 6  | 2 | -0.2 | 0.8  |
| 1  | 4 | 0 | 5  | 2 | -0.2 | 0.8  |
| 16 | 1 | 4 | 21 | 4 | -0.4 | 0.67 |
| 0  | 9 | 4 | 13 | 3 | -0.3 | 0.9  |

$a_i$ 

|    |    |    |    |   |    |
|----|----|----|----|---|----|
| 20 | 10 | 10 | 10 | 0 | 10 |
|----|----|----|----|---|----|

 $b$

$a_i y_i$ 

|    |     |    |     |   |     |
|----|-----|----|-----|---|-----|
| 20 | -10 | 10 | -10 | 0 | -10 |
|----|-----|----|-----|---|-----|

|  |  |
|--|--|
|  |  |
|--|--|

sign 

|  |  |
|--|--|
|  |  |
|--|--|

# SVM

Linear

| $y_i$ |       | $x_i$ |   | $x'_1$ | $x'_2$ |
|-------|-------|-------|---|--------|--------|
| 1     | $x_1$ | 1     | 2 | 1      | 2      |
| -1    | $x_2$ | 0     | 4 | 3      | 1      |
| 1     | $x_3$ | 0     | 3 | 0      | 1      |
| -1    | $x_4$ | 0     | 3 | 1      | 1      |
| 1     | $x_5$ | 5     | 0 | 3      | 1      |
| -1    | $x_6$ | 1     | 4 | 3      | 1      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$  2 0 1 1 1 1  $b$   
 $a_i y_i$  2 0 1 -1 1 -1  $2 \rightarrow 17$   $5$   
 sign + +

Decision Boundary

$$b + \sum a_i y_i K$$

RBF

| $x_i$ | $x'_j$ |   |
|-------|--------|---|
| 1     | 2      | 2 |
| 0     | 4      | 4 |
| 0     | 3      | 3 |

| $-  x - x'  ^2$ | $e^{\square}$ |
|-----------------|---------------|
| -0.5            | 0.6           |
| -0.4            | 0.67          |
| -0.3            | 0.7           |
| -0.2            | 0.8           |
| -0.1            | 0.9           |

|   |
|---|
| 1 |
| 1 |
| 1 |

|   |   |   |   |    |   |    |   |      |     |
|---|---|---|---|----|---|----|---|------|-----|
| 1 | 2 | 1 | 1 | 4  | 4 | 9  | 3 | -0.3 | 0.7 |
| 0 | 4 | 3 | 4 | 0  | 0 | 4  | 2 | -0.2 | 0.8 |
| 0 | 3 | 0 | 0 | 1  | 0 | 1  | 1 | -0.1 | 0.9 |
| 0 | 3 | 1 | 4 | 1  | 4 | 9  | 3 | -0.3 | 0.7 |
| 5 | 0 | 3 | 9 | 16 | 0 | 25 | 5 | -0.5 | 0.6 |
| 1 | 4 | 3 | 1 | 0  | 0 | 1  | 1 | -0.1 | 0.9 |

|    |   |   |    |   |      |      |
|----|---|---|----|---|------|------|
| 0  | 1 | 0 | 1  | 1 | -0.1 | 0.9  |
| 1  | 9 | 4 | 14 | 3 | -0.3 | 0.7  |
| 1  | 4 | 1 | 6  | 2 | -0.2 | 0.8  |
| 1  | 4 | 0 | 5  | 2 | -0.2 | 0.8  |
| 16 | 1 | 4 | 21 | 4 | -0.4 | 0.67 |
| 0  | 9 | 4 | 13 | 3 | -0.3 | 0.9  |

$a_i$  20 10 10 10 0 10  $b$   
 $a_i y_i$  20 -10 10 -10 0 -10  $-1 \rightarrow -2$   $1$   
 sign



# SVM

## Linear

| $y_i$ |       | $x_i$ | $x'_1$ | $x'_2$ |
|-------|-------|-------|--------|--------|
| 1     | $x_1$ | 1 2 1 | 13     | 2      |
| -1    | $x_2$ | 0 4 3 | 25     | 7      |
| 1     | $x_3$ | 0 3 0 | 12     | 3      |
| -1    | $x_4$ | 0 3 1 | 15     | 4      |
| 1     | $x_5$ | 5 0 3 | 19     | 8      |
| -1    | $x_6$ | 1 4 3 | 27     | 8      |

Kernel Matrix

$$K(x_i, x'_j)$$

$a_i$  [2 0 1 1 1 1]  $b$

$a_i y_i$  [2 0 1 -1 1 -1] 2  $\rightarrow$  17 5

sign [ + ] [ + ]

Decision Boundary

$$b + \sum a_i y_i K$$

## RBF

| $x_i$ | $x'_j$ | $L2 \text{ distance }   ^2$ | $(x - x')^2$ | $\sum \sqrt{-\gamma   ^2}$ | $e^{\square}$ |
|-------|--------|-----------------------------|--------------|----------------------------|---------------|
| 1 2 1 | 2      |                             |              | -0.3                       | 0.7           |
| 0 4 3 | 4      |                             |              | -0.2                       | 0.8           |
| 0 3 0 | 3      |                             |              | -0.1                       | 0.9           |
| 0 3 1 |        |                             |              | -0.3                       | 0.7           |
| 5 0 3 |        |                             |              | -0.5                       | 0.6           |
| 1 4 3 |        |                             |              | -0.1                       | 0.9           |

| $-\gamma   ^2$ | $e^{\square}$ |
|----------------|---------------|
| -0.5           | 0.6           |
| -0.4           | 0.67          |
| -0.3           | 0.7           |
| -0.2           | 0.8           |
| -0.1           | 0.9           |

|   |
|---|
| 1 |
| 1 |
| 1 |

|        |    |      |      |
|--------|----|------|------|
| 0 1 0  | 1  | -0.1 | 0.9  |
| 1 9 4  | 14 | -0.3 | 0.7  |
| 1 4 1  | 6  | -0.2 | 0.8  |
| 1 4 0  | 5  | -0.2 | 0.8  |
| 16 1 4 | 21 | -0.4 | 0.67 |
| 0 9 4  | 13 | -0.3 | 0.9  |

$a_i$  [20 10 10 10 0 10]  $b$

$a_i y_i$  [20 -10 10 -10 0 -10] -1  $\rightarrow$  -2

sign [ - ]

$\rightarrow$  1

sign [ + ]